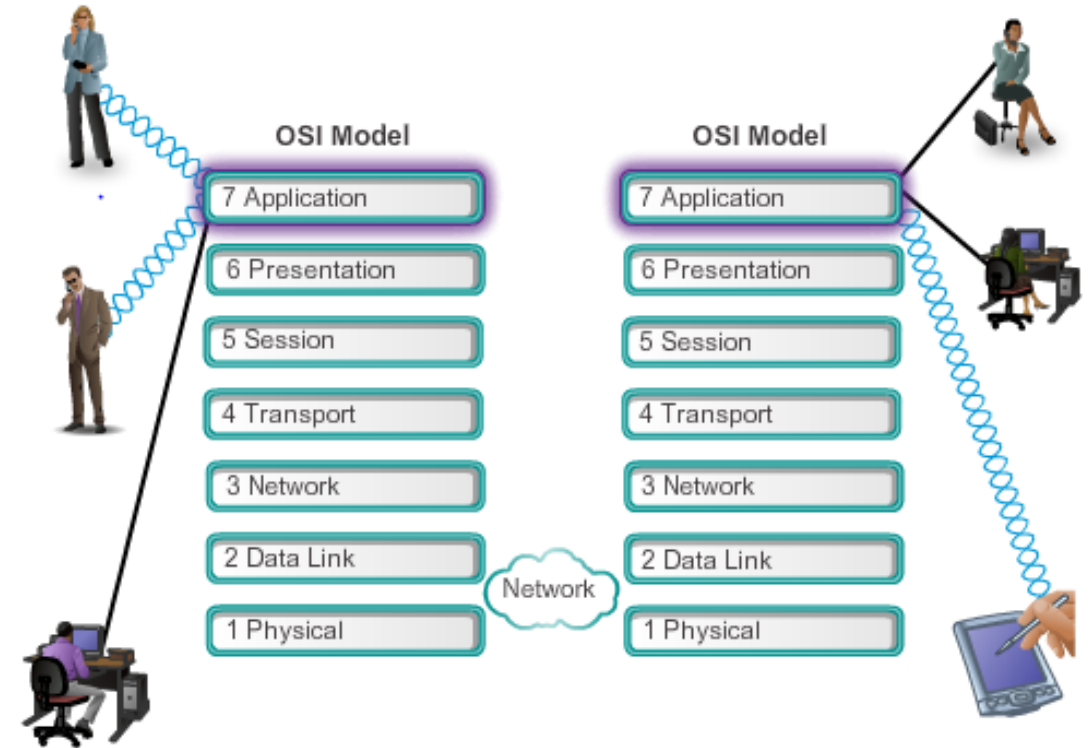


Network Fundamentals

Application layers

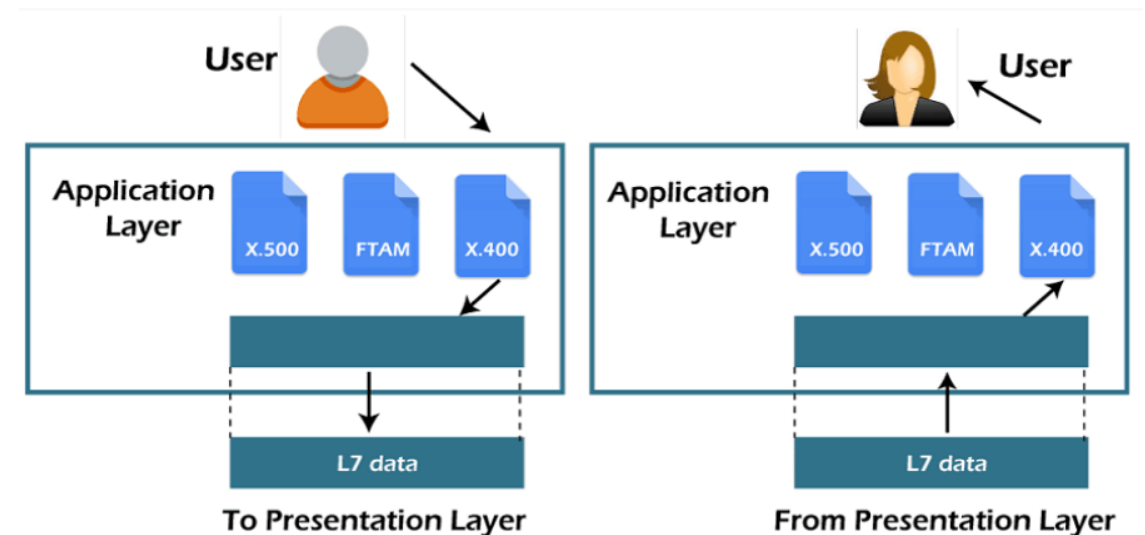
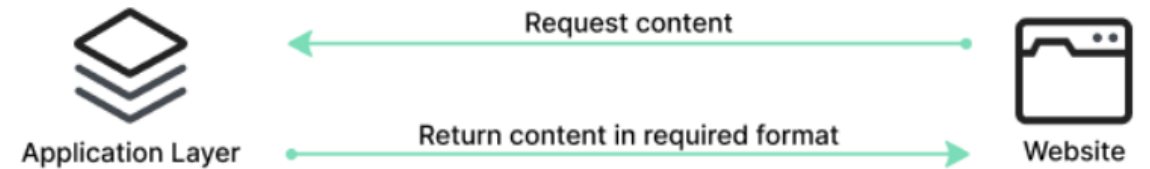
Application Layer

- The Application Layer provides the interface to the network.
- Exchange data between programs on source & destination
- The application layer is closest to the end user



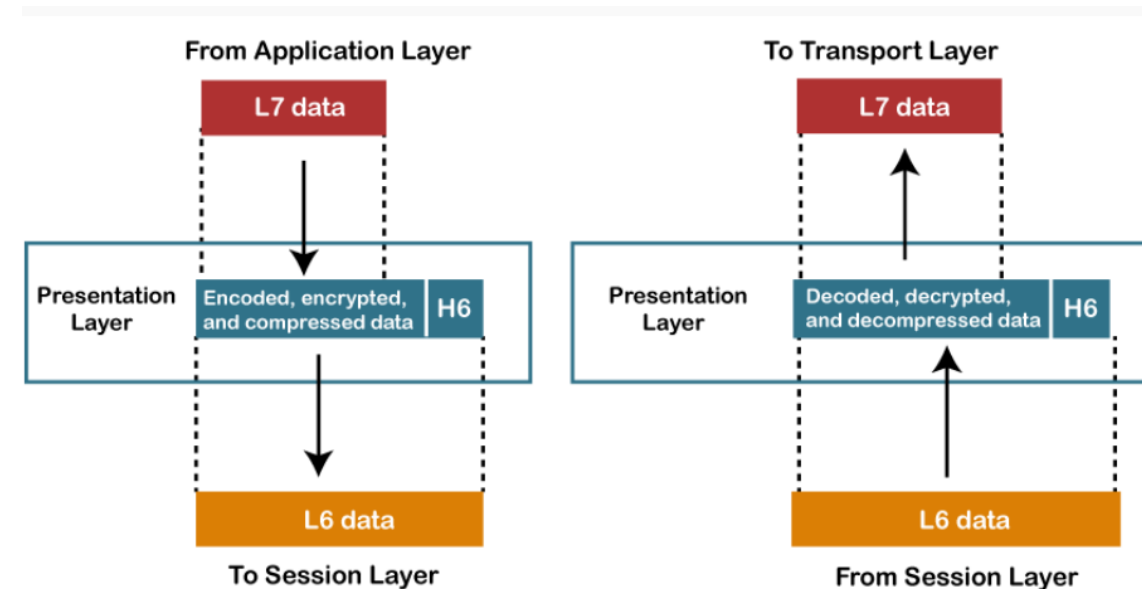
Layer 7 – Application layer

- User interacts with the application layer protocols
- HTTP/HTTPS, FTP/SFTP, DNS, SSH/Telnet, DHCP, SMTP/POP3/IMAP, ...
- Authentication and user access



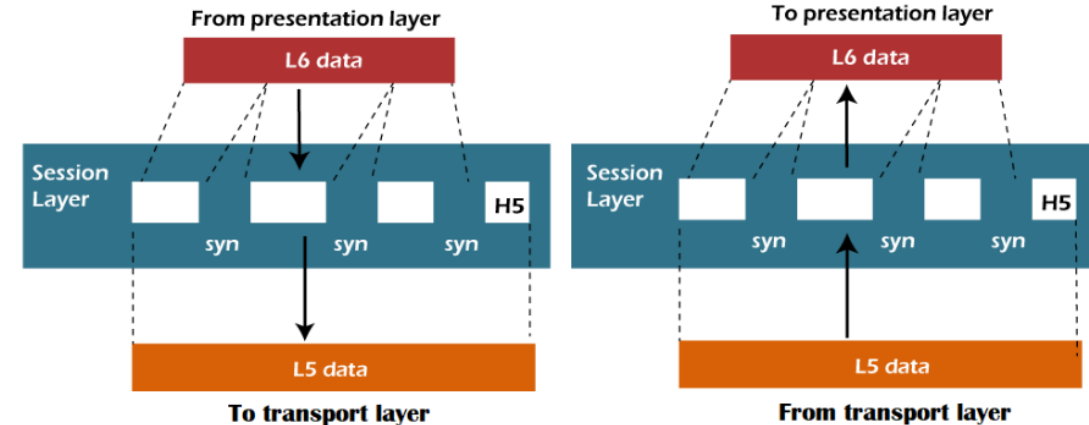
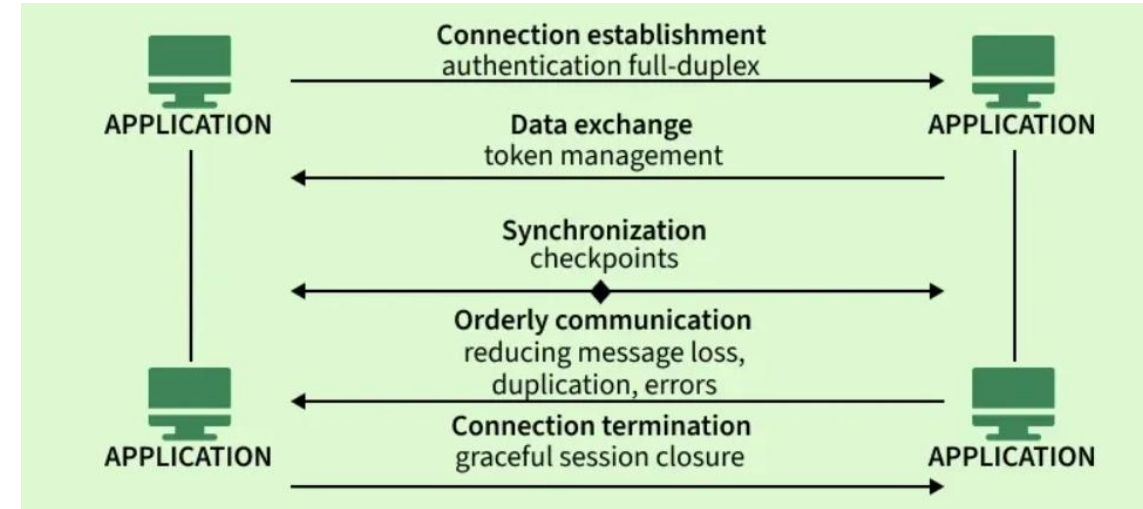
Layer 6 - Presentation Layer

- Data representation
 - json,xml,ASCII, ...
 - JPEG,GIF,MPEG,TIFF,MP3, ...
 - HTML, CSS, JS, ...
- Data coding and translation to the application layer
 - Charset, syntax, ...
- Compression of the data
 - H.264, zip, ...
- Encryption of the data for transmission and the decryption of data upon receipt by the destination
 - SSL/TLS
- Internationalization
 - Dates and times

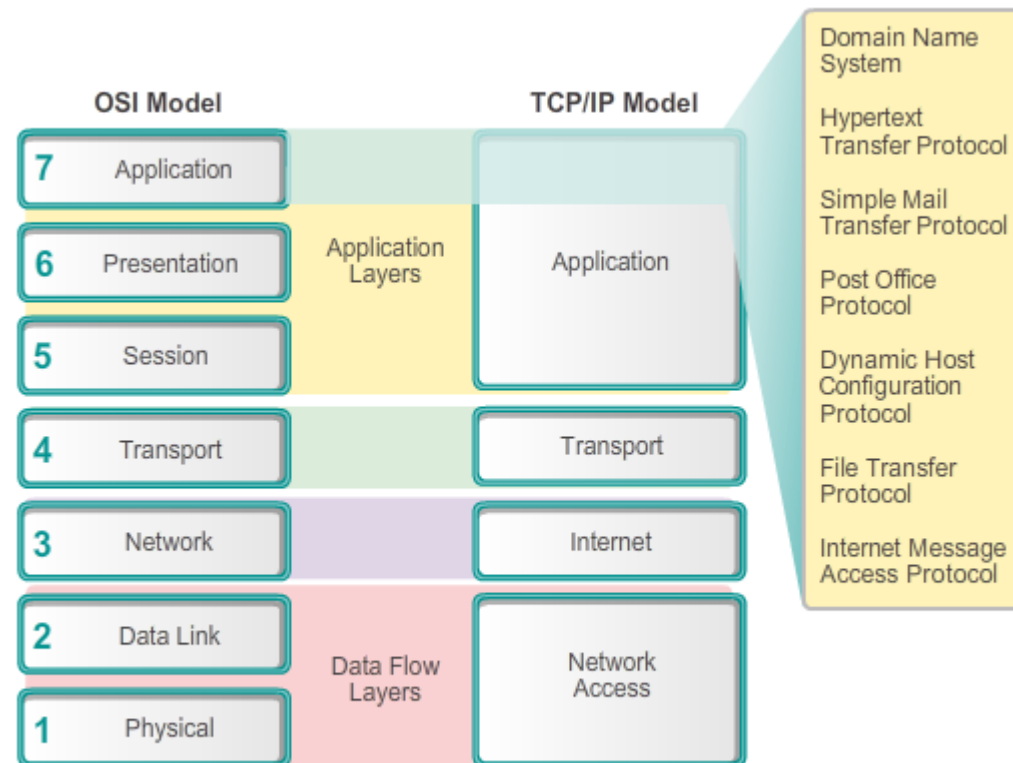


Layer 5 - Session Layer

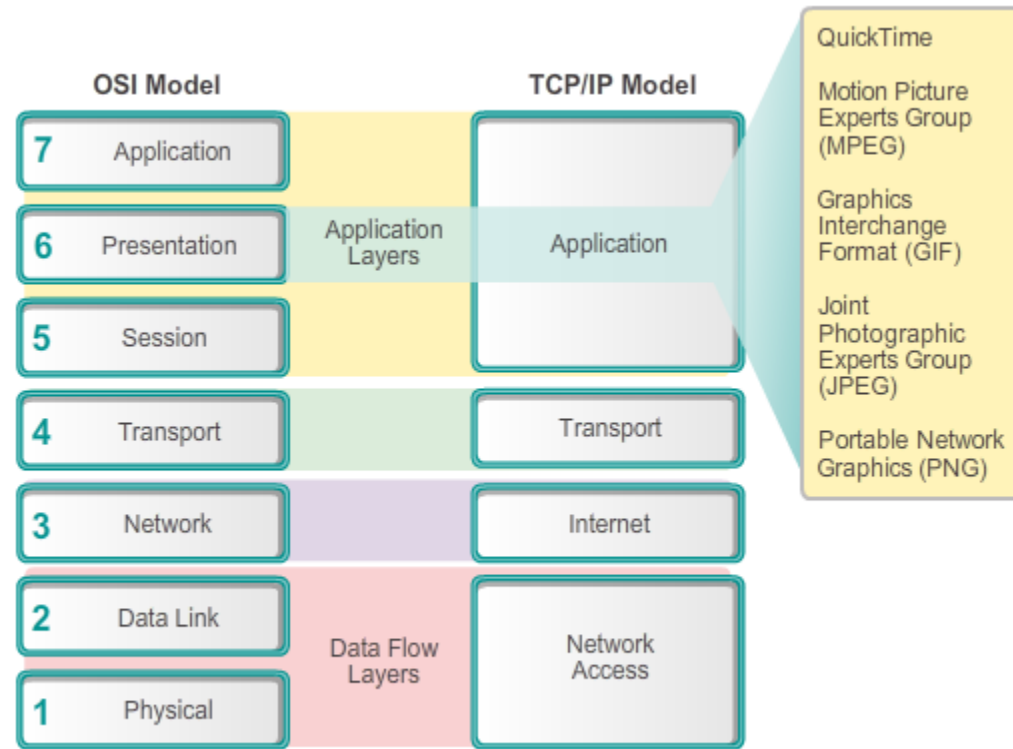
- Create, maintain and terminate dialogs between source and destination applications
- Handles the exchange of information to initiate dialogs, keep them active, and to restart sessions
- Authentication
- Session recovery
- TCP Sessions, API's, Sockets, NetBIOS, RPC



Application Layer Protocols



Presentation Layer



TCP/IP Application Layer Protocols

Network protocols

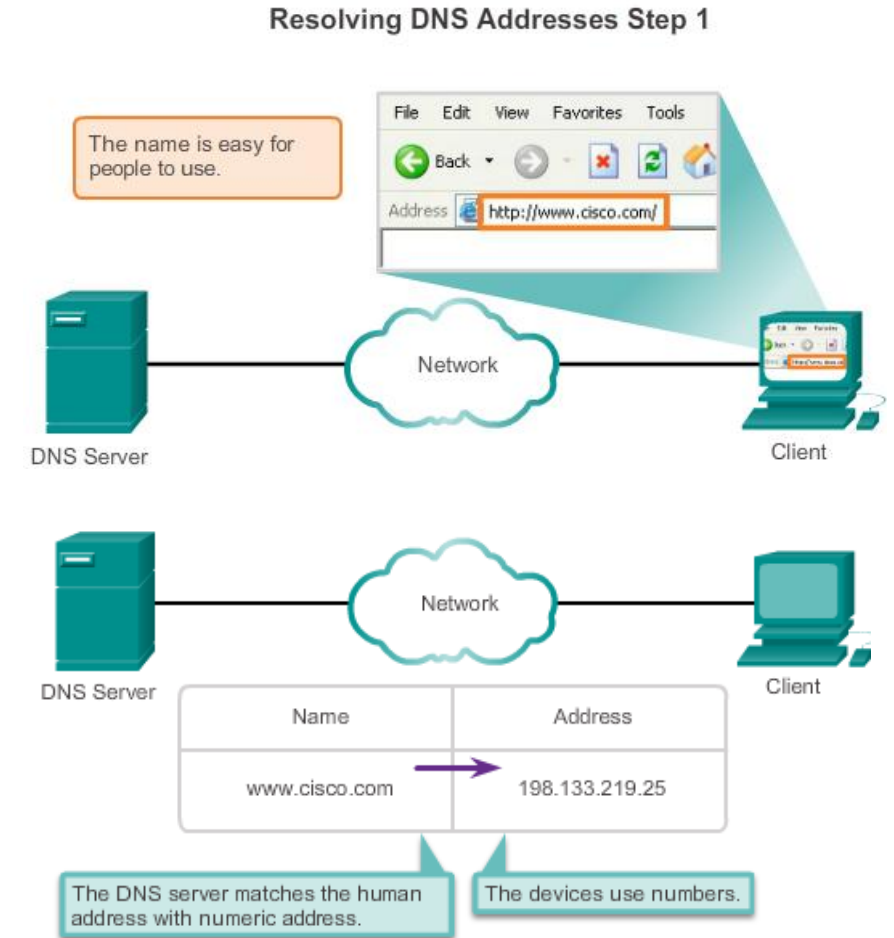
- **Domain Name Service Protocol (DNS)** - used to resolve Internet names to IP addresses
- **Dynamic Host Control Protocol (DHCP)** - used to assign an IP address, subnet mask, default gateway and DNS server to a host
- **Network Time Protocol (NTP)** - synchronizes the clocks on computers and other devices over a network
- **Network Address Translation (NAT)** - map multiple private addresses inside a local network to a public IP address

TCP/IP Application Layer Protocols Applications

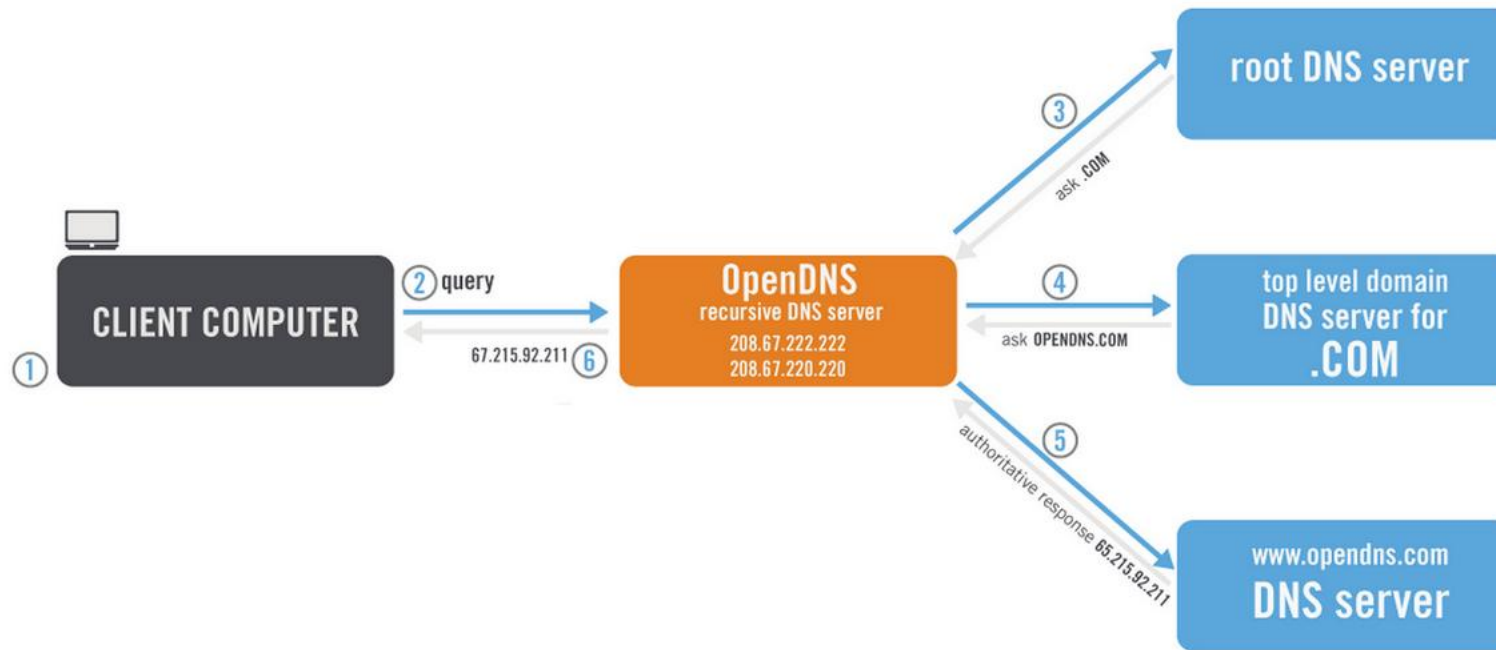
- **Telnet** - a terminal emulation protocol used to provide remote access to servers and networking devices
- **Hypertext Transfer Protocol (HTTP)** - used to transfer files that make up the Web pages of the World Wide Web
- **File Transfer Protocol (FTP)** - used for interactive file transfer between systems
- **Trivial File Transfer Protocol (TFTP)** - used for connectionless active file transfer
- **Simple Mail Transfer Protocol (SMTP)** - used for the transfer of mail messages and attachments
- **Post Office Protocol (POP)** - used by email clients to retrieve email from a remote server
- **Internet Message Access Protocol (IMAP)** – another protocol for email retrieval

Domain Name Service

- On the Internet, domain names, such as `http://www.cisco.com`, are much easier for people to remember than `198.133.219.25`
- The Domain Name System (DNS) translate domain names to ip addresses
- DNS uses a distributed set of servers to resolve the names associated with these numbered addresses.

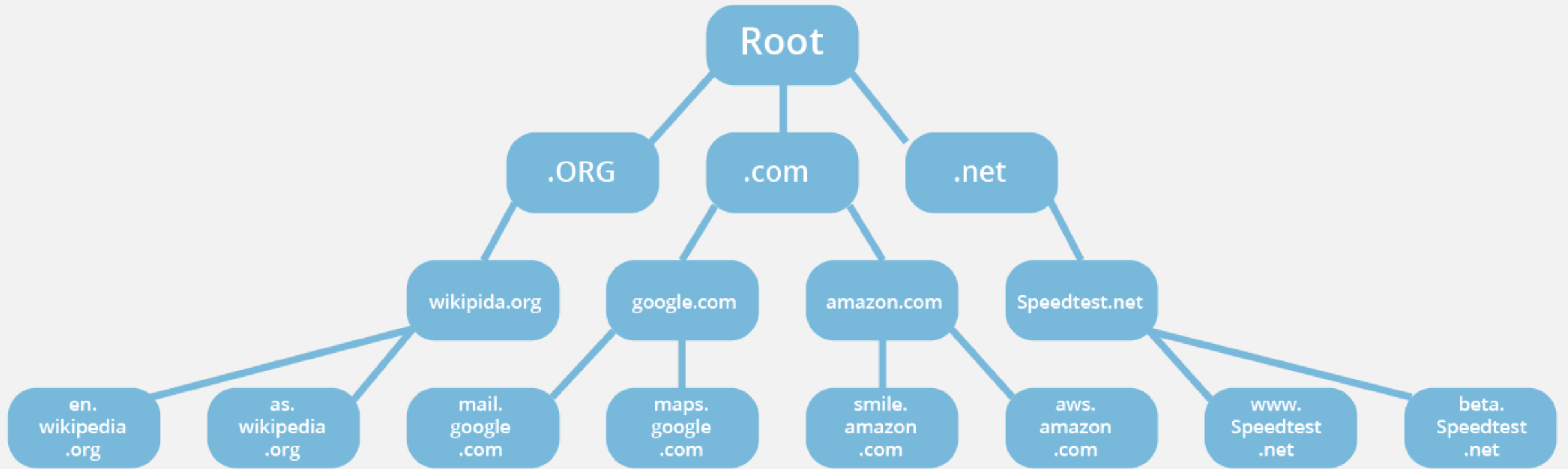


Domain Name Service



This illustration does not take into consideration
DNS caching, web-browser caching, or OS caching.

Domain Name Service



DNS SERVER record types

NS (NameServer): lists the authoritative server for the domain

A (Address record (ipv4)): Maps hostname from the domain to an IPv4 address

AAAA (Address record (ipv6)): Maps hostname from the domain to an IPv6 address

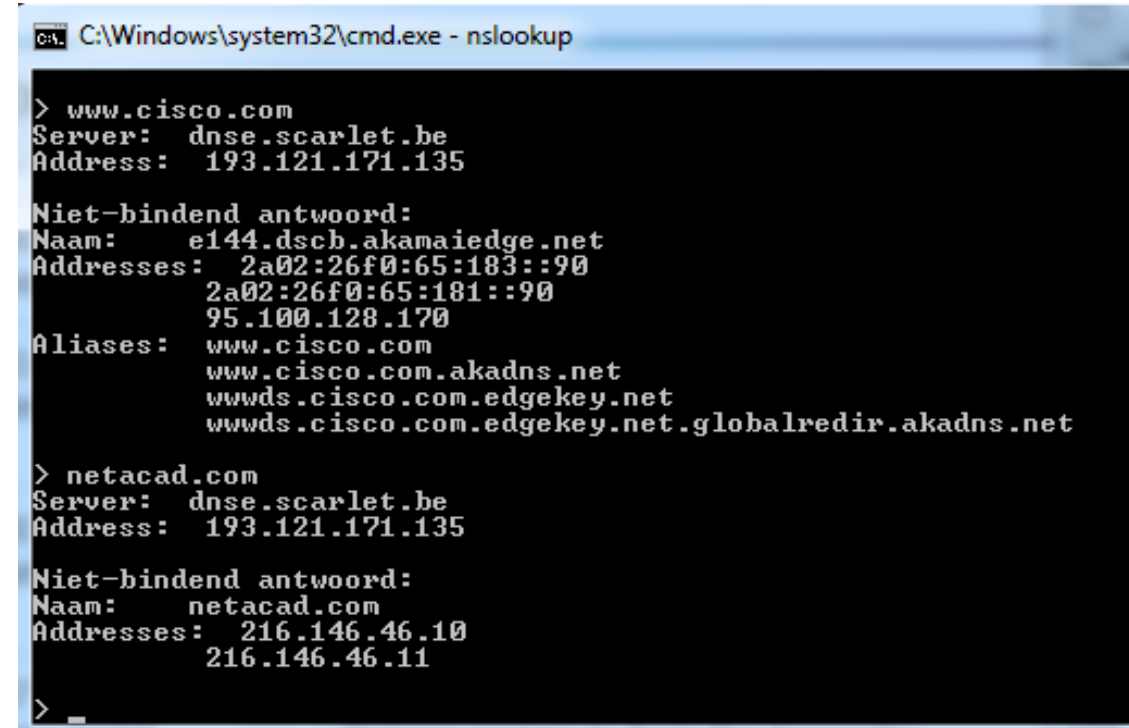
CNAME: Alias to A or AAAA record

MX (Mail exchange record): domain's email server (@domain)

ralphsdomainname.com.	86400	IN	NS	ns1.ralphsdomainname.com.
ralphsdomainname.com.	86400	IN	NS	ns2.ralphsdomainname.com.
ns1	14400	IN	A	109.228.16.61
ns2	14400	IN	A	109.228.18.72
ralphsdomainname.com.	14000	IN	A	109.228.18.72
localhost	14400	IN	A	127.0.0.1
ralphsdomainname.com.	14400	IN	MX	0 ralphsdomainname.com.
mail	14400	IN	CNAME	ralphsdomainname.com.
www	14400	IN	A	109.228.18.72
ftp	14400	IN	CNAME	ralphsdomainname.com.
webdisk	14400	IN	A	109.228.16.61

nslookup

- Operating system utility called **nslookup** allows the user to manually query the name servers to resolve a given host name
- Utility can be used to troubleshoot name resolution issues and to verify the current status of the name servers



```
C:\Windows\system32\cmd.exe - nslookup

> www.cisco.com
Server:  dnse.scarlet.be
Address: 193.121.171.135

Niet-bindend antwoord:
Naam:    e144.dsch.akamaiedge.net
Addresses: 2a02:26f0:65:183::90
           2a02:26f0:65:181::90
           95.100.128.170
Aliases:  www.cisco.com
           www.cisco.com.akadns.net
           wwwds.cisco.com.edgekey.net
           wwwds.cisco.com.edgekey.net.globalredir.akadns.net

> netacad.com
Server:  dnse.scarlet.be
Address: 193.121.171.135

Niet-bindend antwoord:
Naam:    netacad.com
Addresses: 216.146.46.10
           216.146.46.11

> _
```

Dynamic Host Configuration Protocol

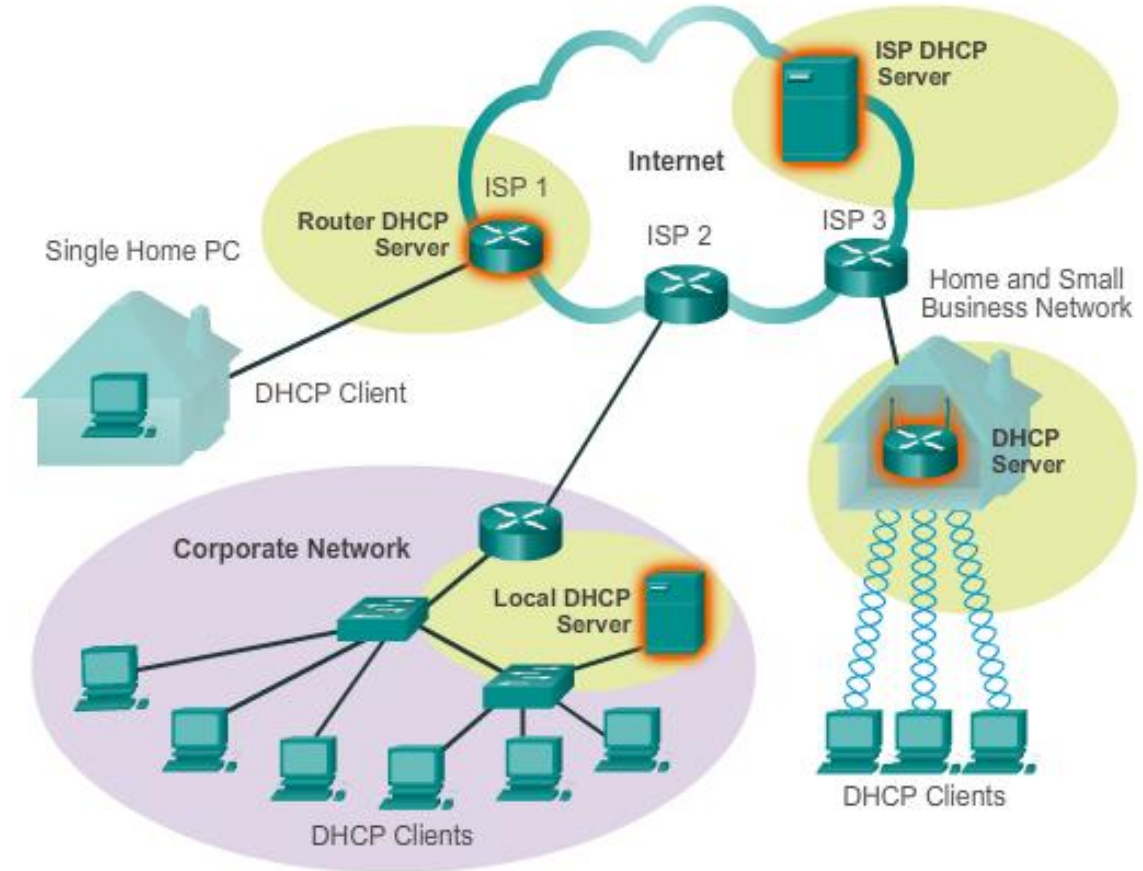
DHCP allows a host to **obtain an IP address** dynamically

DHCP server is contacted and address requested - chooses address from a configured range of addresses called a **pool** and **"leases"** it to the host for a set period

DHCP used for general purpose hosts such as end user devices, and static addressing is used for network devices such as gateways, switches, servers and printers

DHCP server can provide clients with additional **options**

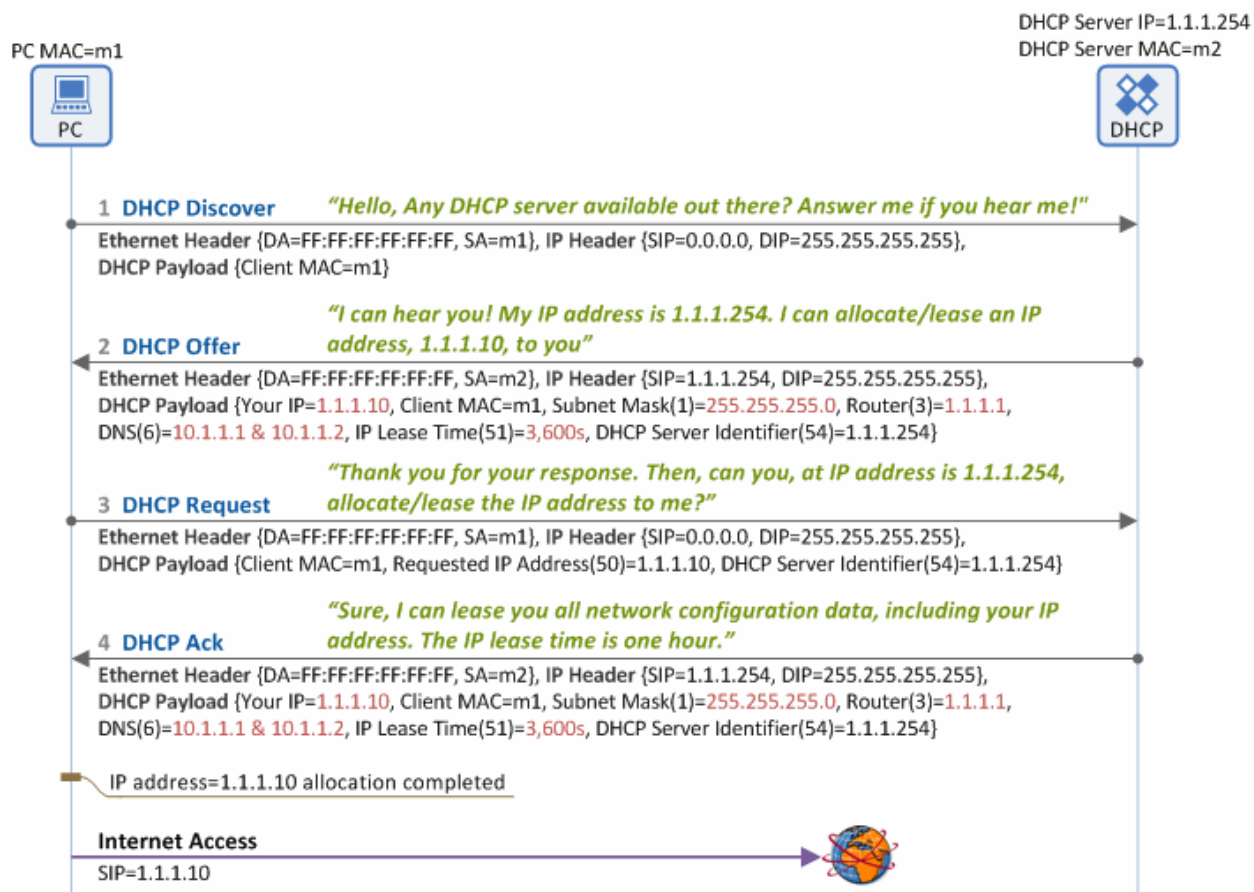
- Default gateway
- DNS server
- NTP server
- TFTP server
- ...



DHCP on Cisco

```
Router(config-if)#ip address 192.168.0.1 255.255.255.0
Router(config)#ip dhcp pool Pool1
Router(dhcp-config)#network 192.168.0.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.0.1
Router(dhcp-config)#dns-server 192.168.0.1
Router(dhcp-config)#exit
Router(config)#ip dhcp exclude-address 192.168.0.0 192.168.0.99
```


DHCP wireshark



No.	Time	Source	Destination	Protocol	Length	Info
28	3.1168...	0.0.0.0	255.255.25...	DHCP	342	DHCP Discover - Transaction ID 0x37e6433e
29	3.1337...	0.0.0.0	255.255.25...	DHCP	342	DHCP Discover - Transaction ID 0x976afb27
52	4.1576...	10.0.0.1	255.255.25...	DHCP	342	DHCP Offer - Transaction ID 0x976afb27
53	4.1595...	0.0.0.0	255.255.25...	DHCP	380	DHCP Request - Transaction ID 0x976afb27
54	4.1720...	10.0.0.1	255.255.25...	DHCP	342	DHCP ACK - Transaction ID 0x976afb27
403	5.5169...	10.0.14.0	10.0.0.1	DHCP	326	DHCP Inform - Transaction ID 0xb754fed
408	5.5315...	10.0.0.1	10.0.14.0	DHCP	342	DHCP ACK - Transaction ID 0xb754fed

<

> Frame 52: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{EF2...}

> Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)

> Internet Protocol Version 4, Src: 10.0.0.1, Dst: 255.255.255.255

> User Datagram Protocol, Src Port: 67, Dst Port: 68

> Dynamic Host Configuration Protocol (Offer)

Message type: Boot Reply (2)

Hardware type: Ethernet (0x01)

Hardware address length: 6

Hops: 0

Transaction ID: 0x976afb27

Seconds elapsed: 0

> Bootp flags: 0x8000, Broadcast flag (Broadcast)

Client IP address: 0.0.0.0

Your (client) IP address: 10.0.14.0

Next server IP address: 0.0.0.0

Relay agent IP address: 0.0.0.0

Client MAC address: Intel_9d:37:e9 (38:87:d5:9d:37:e9)

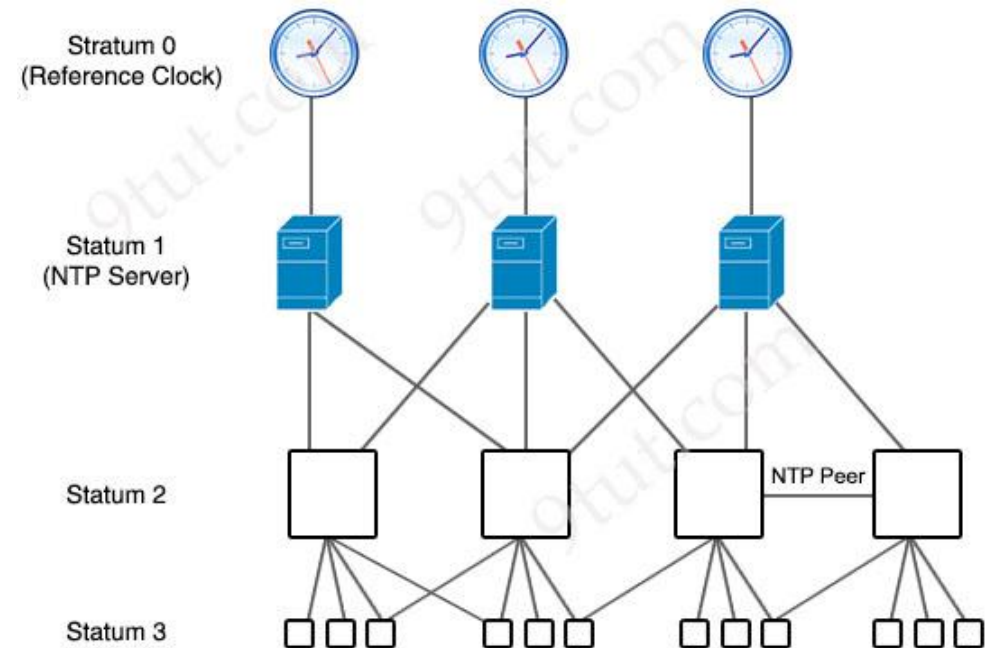
Client hardware address padding: 00000000000000000000

Server host name not given

Boot file name not given

Network Time Protocol (NTP)

- Network devices have an internal clock for time and date
- Wrong or unsynchronized time can cause problems
 - Logs, certificates, ...
- Devices sync time to NTP server
- Stratum 0 – 15 → clock accuracy
- UCP port 123

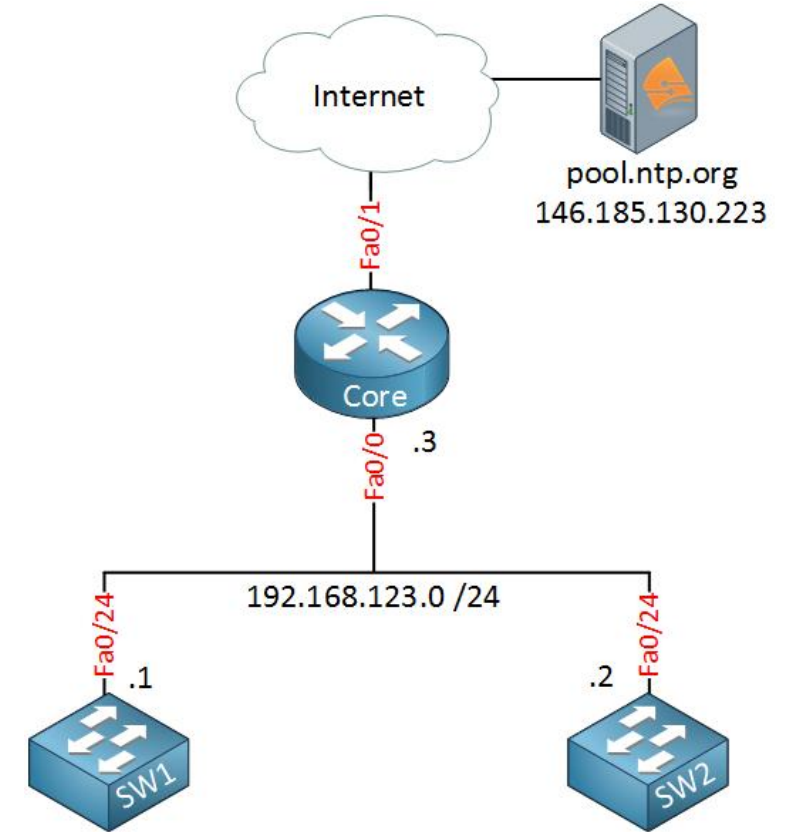


NTP on cisco

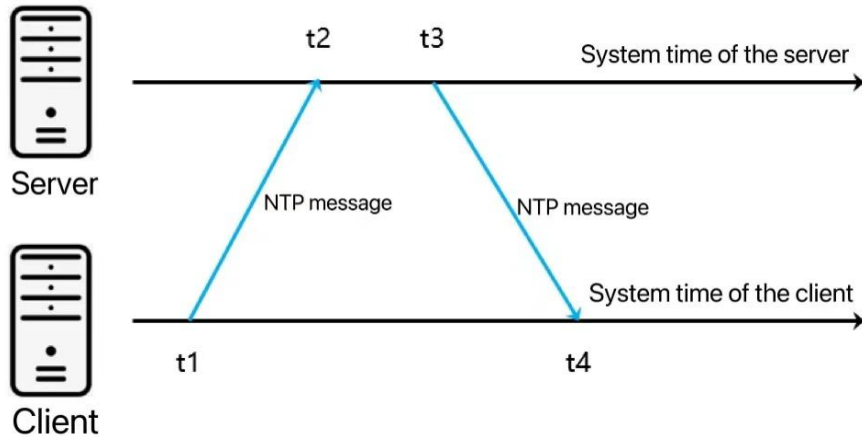
```
CoreRouter(config)#ip name-server 8.8.8.8  
CoreRouter(config)#ntp server pool.ntp.org
```

```
SW1(config)#ntp server 192.168.123.3  
SW2(config)#ntp server 192.168.123.3
```

```
SW1(config)#ntp peer 192.168.123.2  
SW2(config)#ntp peer 192.168.123.1
```



NTP Wireshark



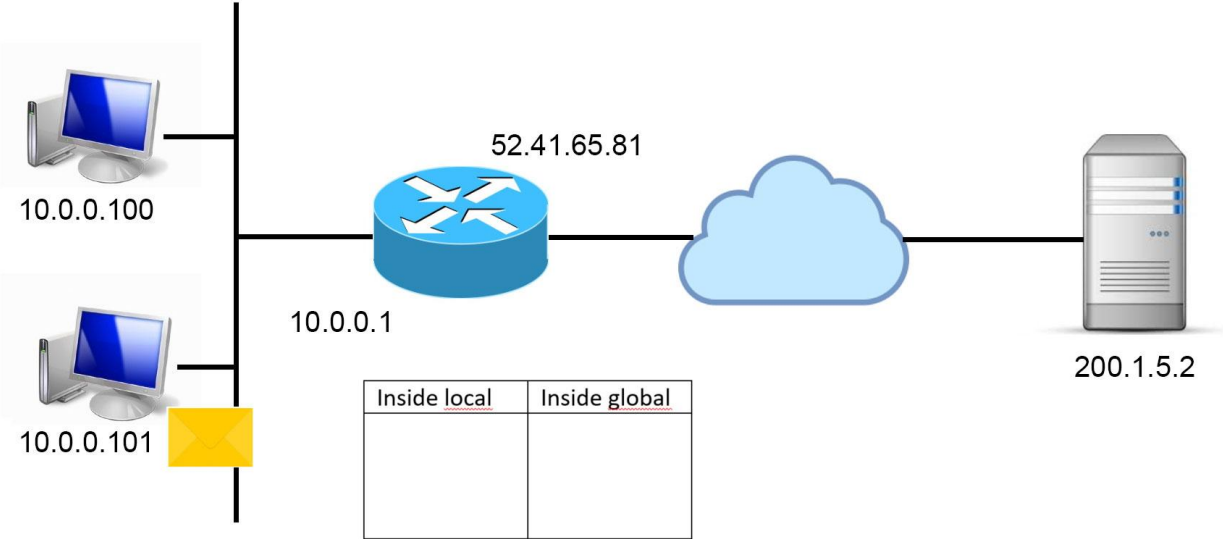
No.	Time	Source	Destination	Protocol	Length	Info
78...	85.640034	10.0.14.0	20.101.57.9	NTP	90	NTP Version 3, client
78...	85.663228	20.101.57.9	10.0.14.0	NTP	90	NTP Version 3, server

```
> Frame 7837: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface \Device\NPF_{EF24EE...}
> Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
> Internet Protocol Version 4, Src: 20.101.57.9, Dst: 10.0.14.0
> User Datagram Protocol, Src Port: 123, Dst Port: 123
> Network Time Protocol (NTP Version 3, server)
  > Flags: 0x1c, Leap Indicator: no warning, Version number: NTP Version 3, Mode: server
    [Request In: 7836]
    [Delta Time: 0.023194000 seconds]
    Peer Clock Stratum: secondary reference (3)
    Peer Polling Interval: 17 (131072 seconds)
    Peer Clock Precision: -23 (0,000000119 seconds)
    Root Delay: 0,001251 seconds
    Root Dispersion: 0,023712 seconds
    Reference ID: 10.0.6.22
    Reference Timestamp: May 11, 2025 18:11:40.744423499 UTC
    Origin Timestamp: May 11, 2025 18:12:55.891617299 UTC
    Receive Timestamp: May 11, 2025 18:12:55.353420999 UTC
    Transmit Timestamp: May 11, 2025 18:12:55.353424599 UTC
```

Network Address Translation (NAT)

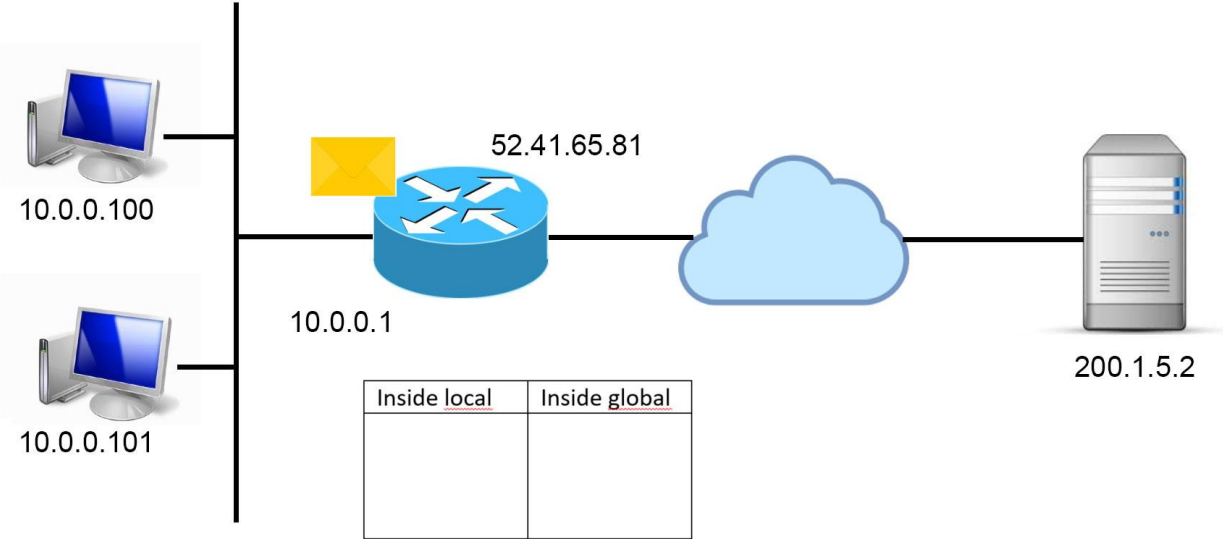
- Router **shares 1 public ip** address with all devices in local network
- Router **replaces the private source ip address and port number** with the public ip address and a unique port number in the packet header
- Router **stores** the private ip and portnumber in a table.
- Router examines the destination ip and port number of incoming packets from the internet and **replaces the public destination ip and portnumber** with the private ip and port number from the table.

Network Address Translation (NAT)



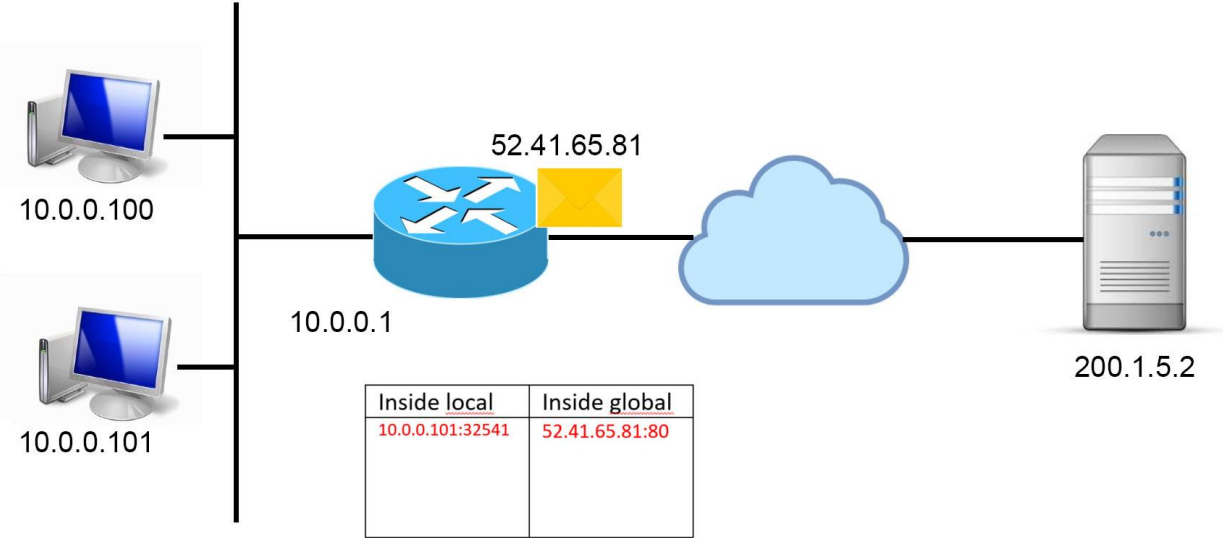
Source IP	Destination IP	Source Port	Destination Port	Data
10.0.0.101	200.1.5.2	32541	80	HTTP GET

Network Address Translation (NAT)



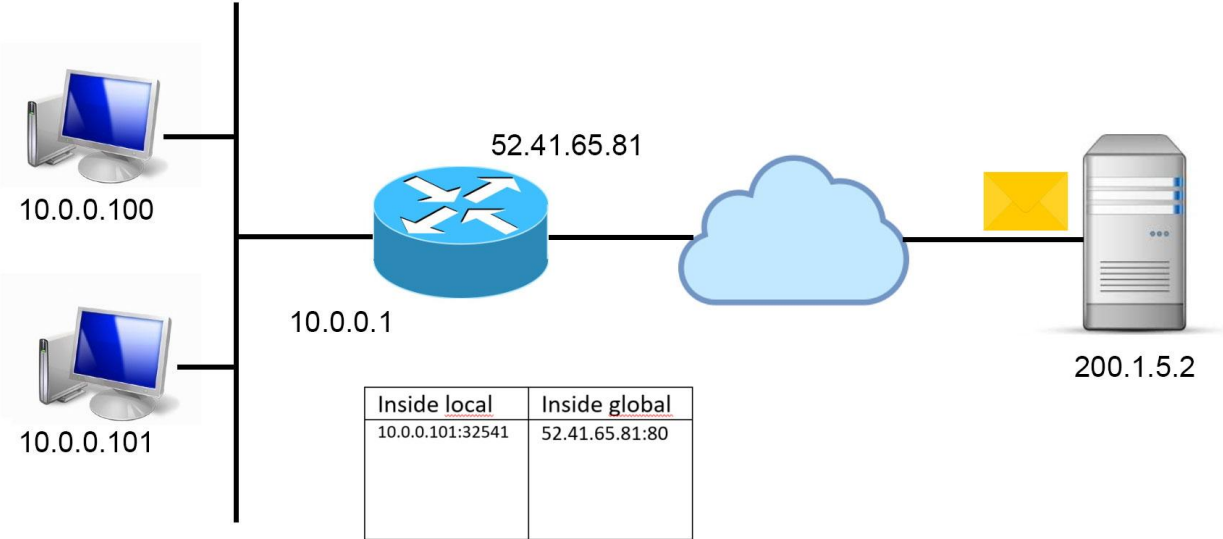
Source IP	Destination IP	Source Port	Destination Port	Data
10.0.0.101	200.1.5.2	32541	80	HTTP GET

Network Address Translation (NAT)



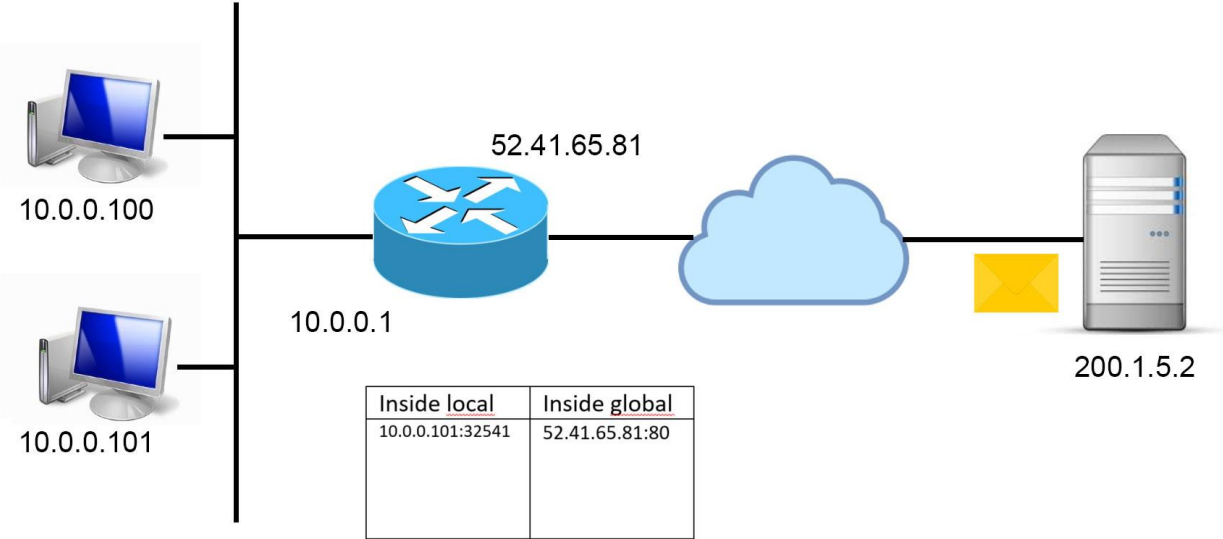
Source IP	Destination IP	Source Port	Destination Port	Data
52.41.65.81	200.1.5.2	11223	80	HTTP GET

Network Address Translation (NAT)



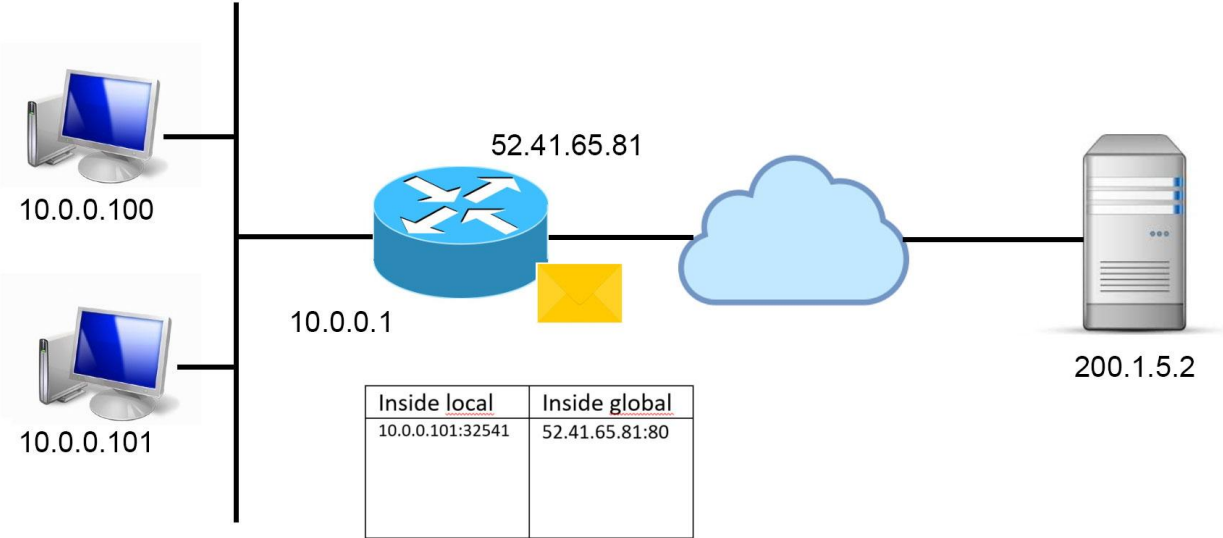
Source IP	Destination IP	Source Port	Destination Port	Data
52.41.65.81	200.1.5.2	11223	80	HTTP GET

Network Address Translation (NAT)



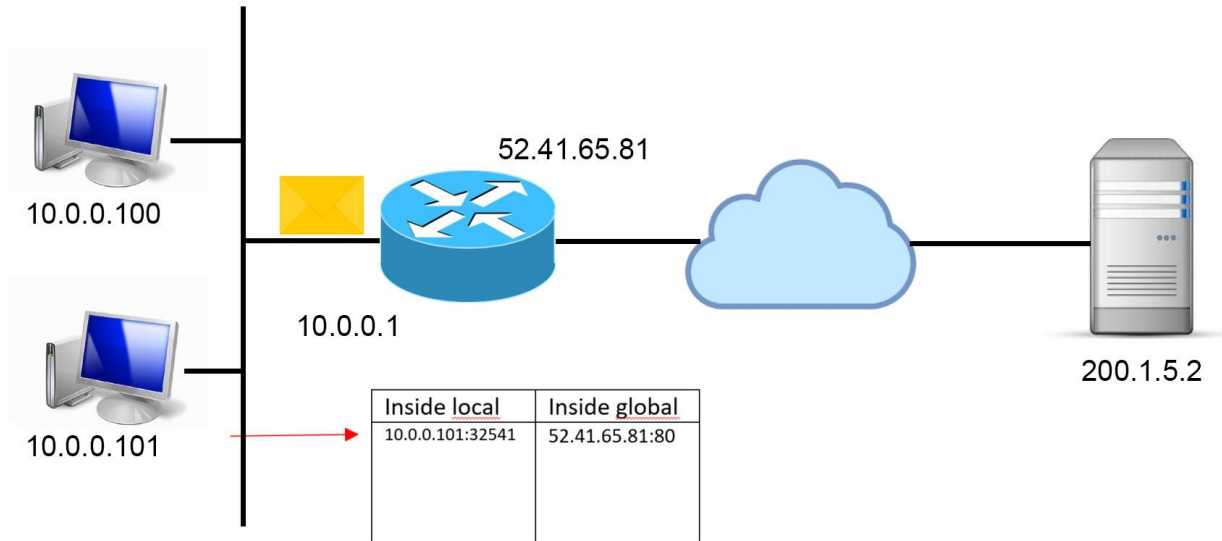
Source IP	Destination IP	Source Port	Destination Port	Data
200.1.5.2	52.41.65.81	80	11223	HTTP Response

Network Address Translation (NAT)



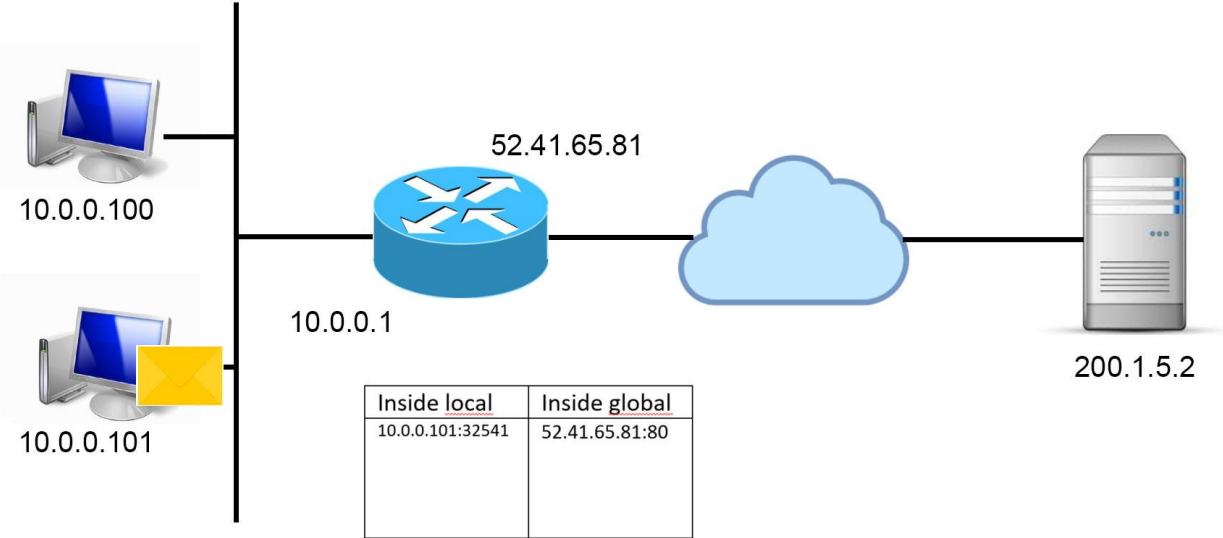
Source IP	Destination IP	Source Port	Destination Port	Data
200.1.5.2	52.41.65.81	80	11223	HTTP Response

Network Address Translation (NAT)



Source IP	Destination IP	Source Port	Destination Port	Data
200.1.5.2	10.0.0.101	80	32541	HTTP Response

Network Address Translation (NAT)

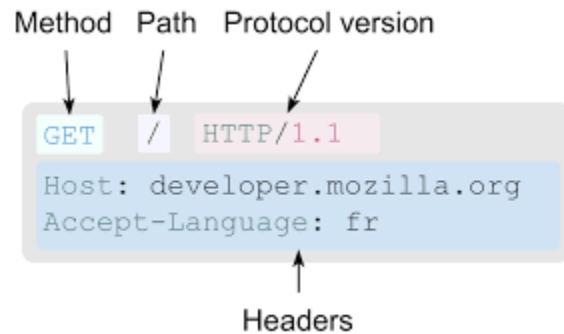


Source IP	Destination IP	Source Port	Destination Port	Data
200.1.5.2	10.0.0.101	80	32541	HTTP Response

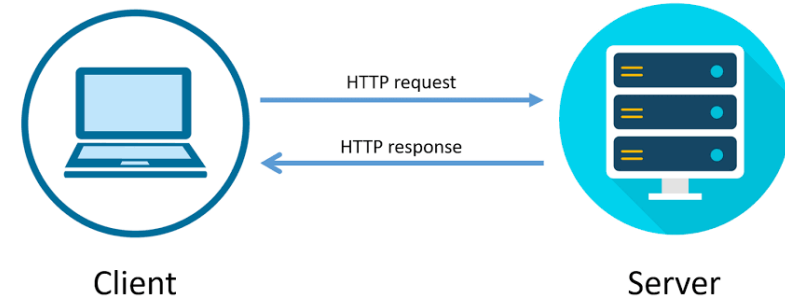
Hypertext Transfer Protocol (http)

TCP port 80

Request → response



CRUD	HTTP	REST
Create	POST	/api/movie
Read	GET	/api/movie/{id}
Update	PUT	/api/movie
Delete	DELETE	/api/movie/{id}



Sr. No.	Code and Description
1.	1xx: Informational It means the request has been received and the process is continuing.
2.	2xx: Success It means the action was successfully received, understood and accepted.
3.	3xx: Redirection It means further action must be taken in order to complete the request.
4.	4xx: Client Error It means the request contains incorrect syntax or cannot be fulfilled.
5.	5xx: Server Error It means the server failed to fulfill an apparently valid request.

http response codes

1xx Informational	
100	Continue
101	Switching protocols
102	Processing
103	Early Hints
3xx Redirection	
301	Moved Permanently
302	Found
303	See Other
304	Not Modified
2xx Successful	
200	OK
201	Created
202	Accepted
203	Non-Authoritative
204	No Content
205	Reset Content
206	Partial Content
207	Multi-Status
208	Already Reported
4xx Client Error	
400	Bad Request
401	Unauthorized
402	Payment required
403	Forbidden
404	Not Found
405	Method Not Allowed
406	Not Acceptable
409	Conflict
413	Payload Too Large
429	Too Many Requests
5xx Server Error	
500	Internal Server Error
501	Not Implemented
502	Bad Gateway
503	Service Unavailable
504	Gateway Timeout
505	HTTP Version Not Supported
506	Variant Also Negotiates
507	Insufficient Storage
508	Loop Detected
510	Not Extended
511	Network Authentication Required

http wireshark

http						
No.	Time	Source	Destination	Protocol	Length	Info
→ 202	5.1039...	10.0.14.0	104.117.77...	HTTP	165	GET /connecttest.txt HTTP/1.1
← 214	5.1353...	104.117.77...	10.0.14.0	HTTP	241	HTTP/1.1 200 OK (text/plain)
<						
Frame 202: 165 bytes on wire (1320 bits), 165 bytes captured (1320 bits) on interface \Device\NPF_{						
Ethernet II, Src: Intel_9d:37:e9 (38:87:d5:9d:37:e9), Dst: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9)						
Internet Protocol Version 4, Src: 10.0.14.0, Dst: 104.117.77.161						
Transmission Control Protocol, Src Port: 41402, Dst Port: 80, Seq: 1, Ack: 1, Len: 111						
Hypertext Transfer Protocol						
GET /connecttest.txt HTTP/1.1\r\n						
Request Method: GET						
Request URI: /connecttest.txt						
Request Version: HTTP/1.1						
Connection: Close\r\n						
User-Agent: Microsoft NCSI\r\n						
Host: www.msftconnecttest.com\r\n						
\r\n						
[Response in frame: 214]						
[Full request URI: http://www.msftconnecttest.com/connecttest.txt]						

https

- TCP port 443
- Encrypted and authenticated data

No.	Time	Source	Destination	Protocol	Length	Info
71...	68.920...	10.0.14.0	151.101.19...	TCP	66	42595 → 443 [SYN] Seq=0 Win=64
71...	68.943...	151.101.19...	10.0.14.0	TCP	66	443 → 42595 [SYN, ACK] Seq=0 A
71...	68.943...	10.0.14.0	151.101.19...	TCP	54	42595 → 443 [ACK] Seq=1 Ack=1
71...	68.944...	10.0.14.0	151.101.19...	TLS...	1777	Client Hello (SNI=thomasmore.b
71...	68.968...	151.101.19...	10.0.14.0	TCP	60	443 → 42595 [ACK] Seq=1 Ack=17
71...	68.971...	151.101.19...	10.0.14.0	TLS...	1510	Server Hello, Change Cipher Sp
71...	68.971...	151.101.19...	10.0.14.0	TCP	1510	443 → 42595 [ACK] Seq=145

Frame 7173: 1777 bytes on wire (14216 bits), 1777 bytes captured (14216 bits)

Ethernet II, Src: Intel_9d:37:e9 (38:87:d5:9d:37:e9), Dst: VMware_fb:a0:c9 (08:00:27:eb:a0:c9)

Internet Protocol Version 4, Src: 10.0.14.0, Dst: 151.101.193.91

Transmission Control Protocol, Src Port: 42595, Dst Port: 443, Seq: 1, Ack: 1,

Transport Layer Security

TLShv1.3 Record Layer: Handshake Protocol: Client Hello

Content Type: Handshake (22)

Version: TLS 1.0 (0x0301)

Length: 1718

Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 1714

Version: TLS 1.2 (0x0303)

Random: 547755fe6d2acdcb0803c715c66fde991ea4baa9cfff6d56934c493947a92f9d5

Session ID Length: 32

Session ID: e8ce510223eb410f5cb719d33f96437bd337b227dc1b9609405c409f7b053

Cipher Suites Length: 32

Cipher Suites (16 suites)

Compression Methods Length: 1

Compression Methods (1 method)

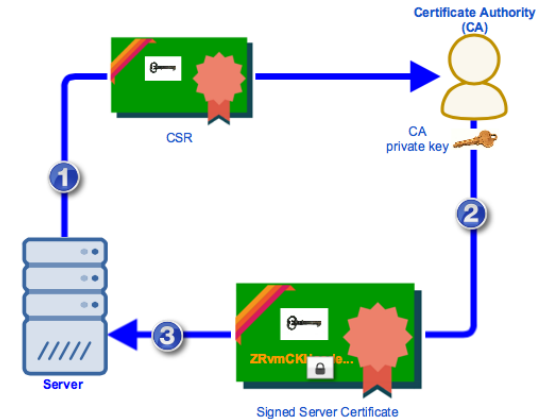
Extensions Length: 1609

Extension: Reserved (GREASE) (len=0)

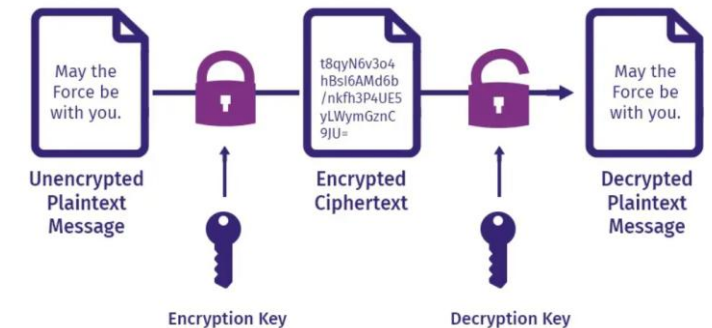
Extension: supported_versions (len=7) TLS 1.3 TLS 1.2

Authentication and Encryption

Authentication: Identity verification using signed certificates

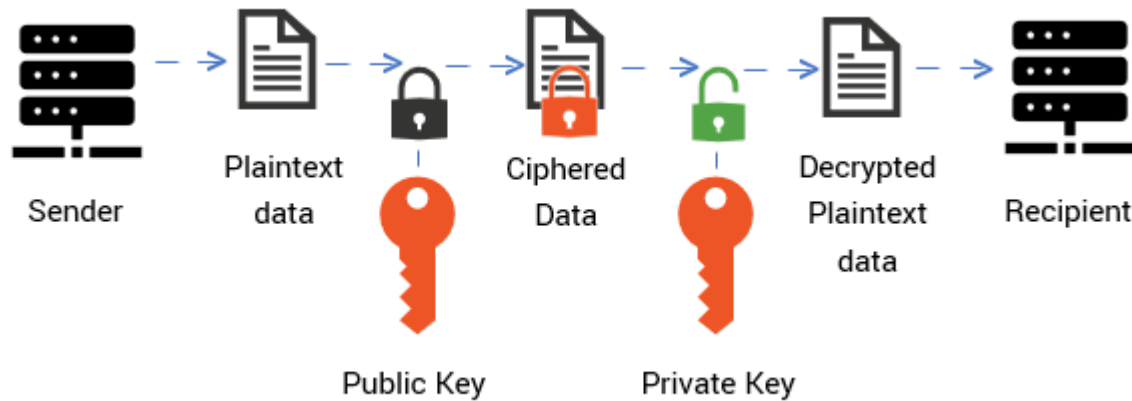


Encryption: Data protection using symmetric session keys



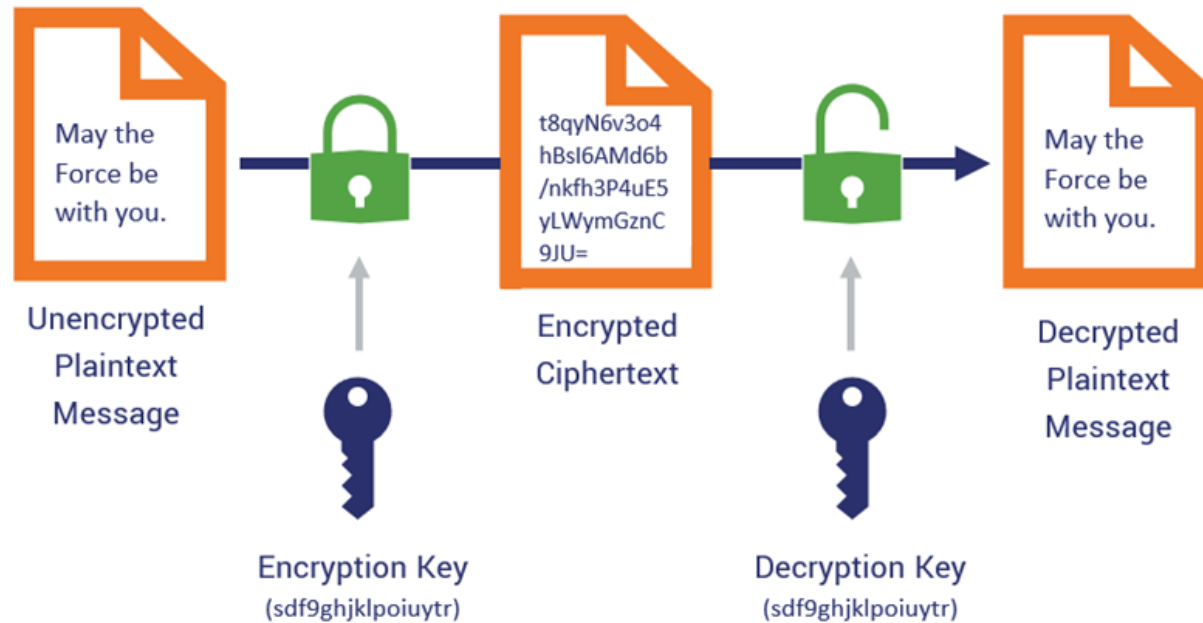
Asymmetric public private keypair

Data encrypted by the private key can only be decrypted by the private key



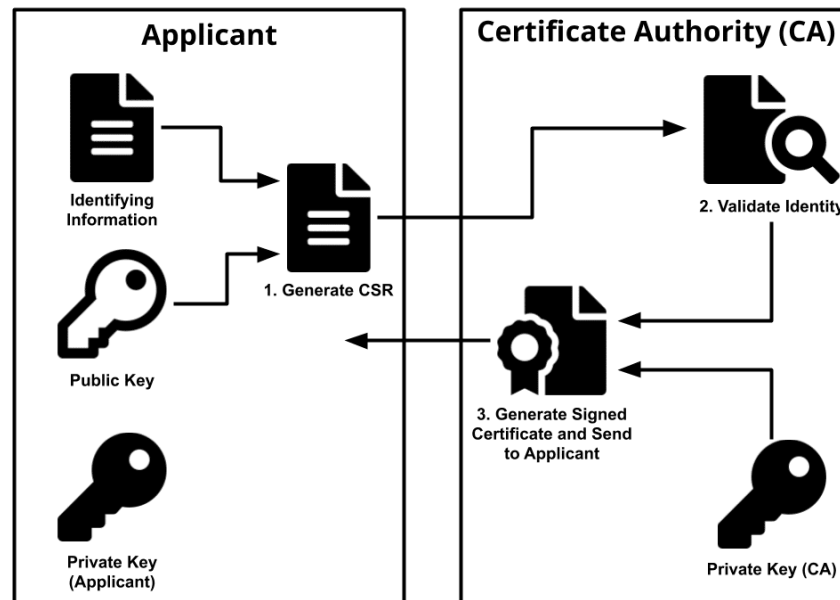
Symmetric session keys

Symmetric keys can both encrypt and decrypt data



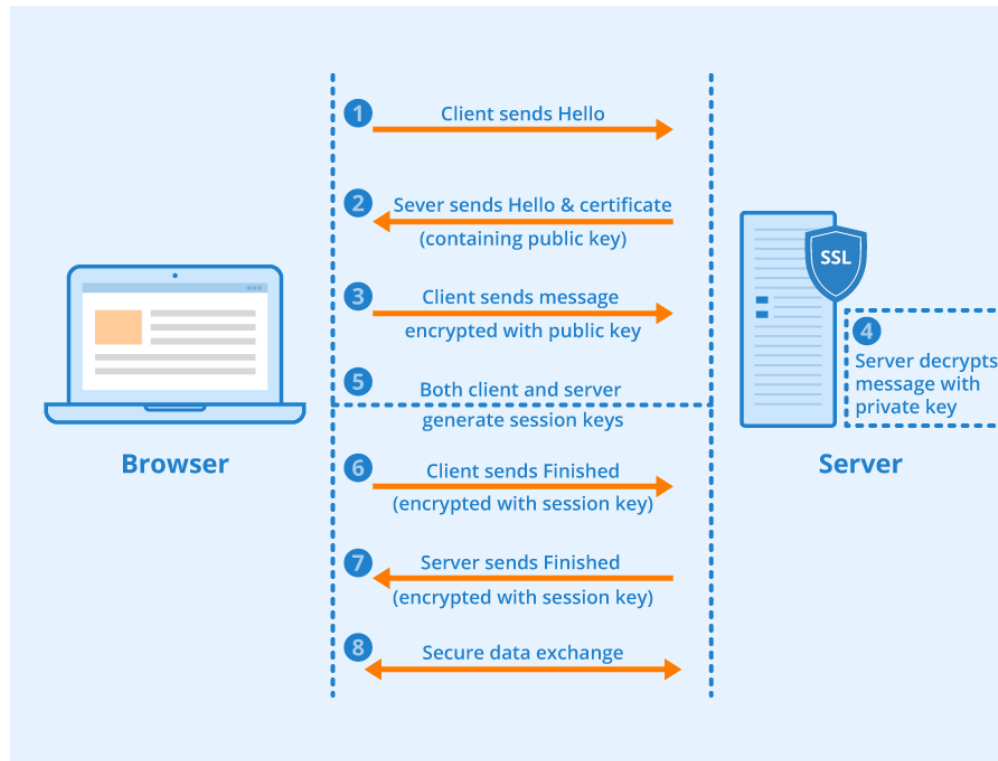
Certificate signature

- Certificate Authority: Limited list of trusted authorities baked into OS
 - Let'sEncrypt, DigiCert, RapidSSL, ...
- Applicant sends identity information and public key to CA
- CA uses applicant public key and CA private key to generate signed certificate
- Certificate is linked to domain name



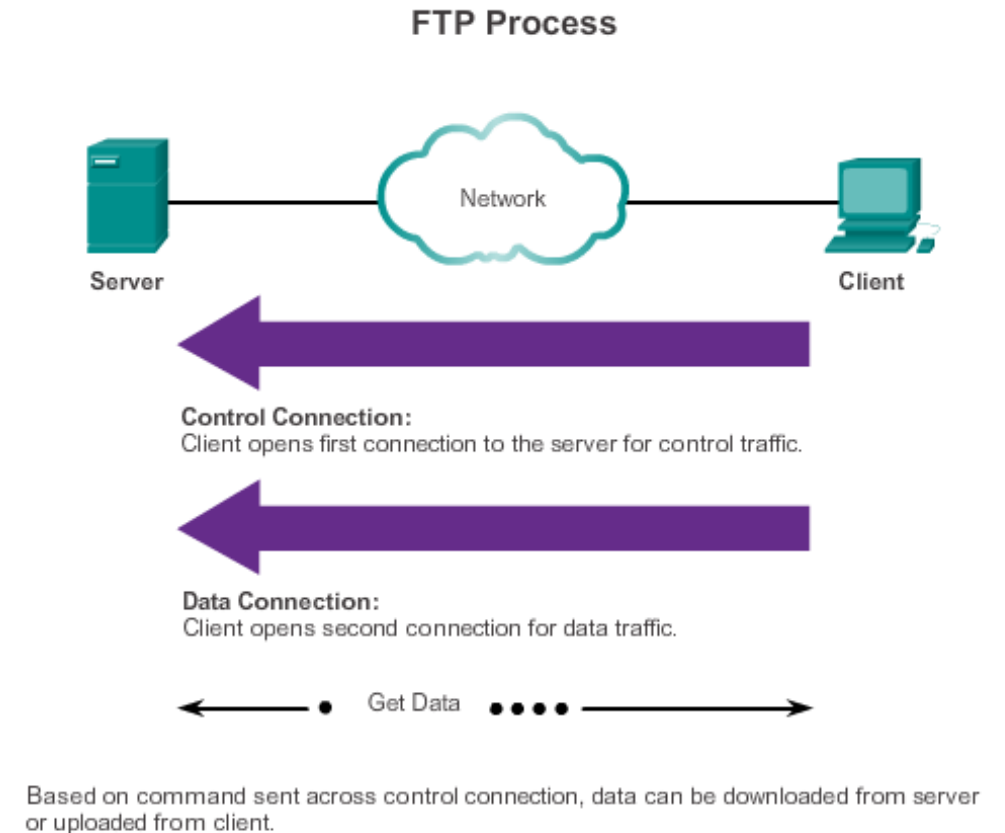
tls handshake process

Using asynchronous keys to share signed certificates and generate session keys which are used to encrypt the session data



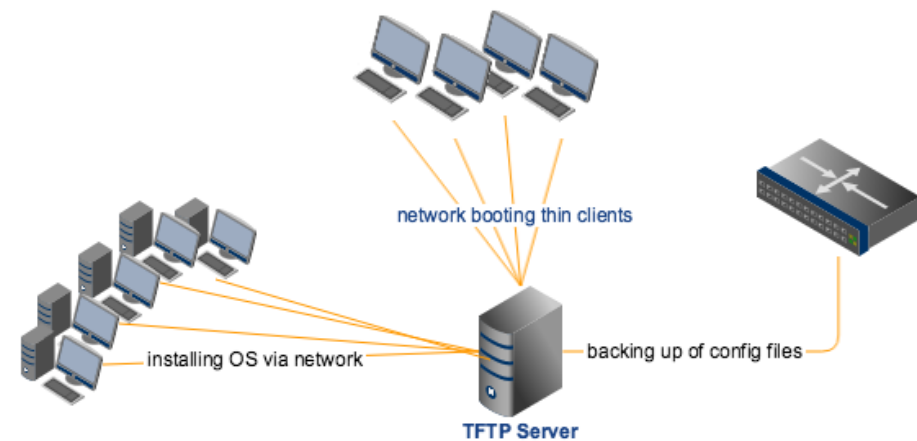
File Transfer Protocol

- FTP allow data transfers between a client and a server
- FTP client is an application that runs on a computer that is used to push and pull data from a server running an FTP daemon
- To successfully transfer data, FTP requires two connections between the client and the server, one for commands and replies **port 21**, the other for the actual data transfer **port 20**



Trivial File transfer protocol

- Simper version of FTP
- Uses UDP on default port 69
- No folder listing
- No security like username/password or encryption
 - Use only on trusted local area networks!



Telnet/SSH

- Send commands from CLI to remote device over network
 - Telnet: plain text, port 23
 - Use only on trusted local area networks!
 - SSH: encrypted, port 22
 - Login using Username/password and/or Private key

